

$$\frac{\partial \theta}{\partial r} = \frac{\partial T}{\partial r} \cdot \frac{r_0}{r} \cdot \frac{\partial \theta}{\partial T}$$

$$= \frac{\partial T}{\partial r} \cdot r_0 \cdot \frac{1}{(T_0 - T_\infty)}$$

$$\frac{\partial T}{\partial r} = \frac{\partial \theta}{\partial r} \cdot \frac{\partial T}{\partial \theta}$$

$$= (T_0 - T_\infty) \frac{\partial \theta}{\partial r}$$

$$\frac{\partial T}{\partial r} = \frac{(T_0 - T_\infty)}{r_0} \cdot \frac{\partial \theta}{\partial r}$$

$$r_0 \frac{\partial T}{\partial r} = r_0 \cdot \frac{(T_0 - T_\infty)}{r_0} \cdot \frac{\partial \theta}{\partial r}$$

$$\frac{\partial}{\partial r} \left(r_0 \frac{\partial T}{\partial r} \right) = \frac{\partial}{\partial r} \left(r_0 \cdot \frac{(T_0 - T_\infty)}{r_0} \cdot \frac{\partial \theta}{\partial r} \right) \cdot \frac{\partial r}{\partial r} \cdot \frac{1}{r_0}$$

$$= \frac{\partial}{\partial r} \left(\lambda \cdot r_0 \cdot \frac{(T_0 - T_\infty)}{r_0} \cdot \frac{\partial \theta}{\partial r} \right)$$

$$= \frac{\partial}{\partial r} \left(\lambda \cdot (T_0 - T_\infty) \cdot \frac{\partial \theta}{\partial r} \right)$$

$$\frac{1}{r_0} \frac{\partial}{\partial r} \left(r_0 \frac{\partial T}{\partial r} \right) = \frac{1}{\lambda r_0} \cdot \frac{\partial}{\partial r} \left[\lambda (T_0 - T_\infty) \frac{\partial \theta}{\partial r} \right]$$

$$= \frac{(T_0 - T_\infty)}{r_0} \cdot \frac{1}{\lambda} \frac{\partial}{\partial r} \left[\lambda \frac{\partial \theta}{\partial r} \right]$$

$$\frac{(T_0 - T_\infty)}{r_0} \cdot \frac{1}{\lambda} \frac{\partial}{\partial r} \left(\lambda \frac{\partial \theta}{\partial r} \right) = \frac{1}{\alpha} \cdot \frac{(T_0 - T_\infty)}{r_0} \frac{\partial \theta}{\partial r}$$

$$\frac{1}{\lambda} \frac{\partial}{\partial r} \left(\lambda \frac{\partial \theta}{\partial r} \right) = \frac{r_0}{\alpha} \frac{\partial \theta}{\partial r}$$

$$\frac{1}{\lambda} \frac{\partial}{\partial r} \left(\lambda \frac{\partial \theta}{\partial r} \right) = \frac{\partial \theta}{\partial r}$$

$$\left[\lambda \frac{\partial \theta}{\partial r} \right]$$

$$\theta = 0, \quad \theta = 1$$

$$\theta = 0, \quad \theta = h r_0$$

$$\theta = \lambda^2, \quad -h \cdot \left(\frac{T_0 - T_\infty}{r_0} \right) \cdot \frac{\partial \theta}{\partial r} = h (T_0 - T_\infty)$$

$$\frac{\partial \theta}{\partial r} + h r_0 = 0$$

Q4)

dual phase lag.

$$\frac{\partial \tau}{\partial T} = -$$

$$q(r, t + \tau_0) = -k \nabla T(r, t + \tau_T)$$

$$q(r, t) + \tau_0 \frac{\partial q(r, t)}{\partial t} = -k \left[\nabla T(r, t) + \tau_T \frac{\partial \nabla T(r, t)}{\partial t} \right]$$

heat eqn.

$$-\nabla \cdot q''$$

$$\frac{1}{\alpha} \nabla^2 T + \tau_T \frac{\partial}{\partial t} (\nabla^2 T) + \frac{1}{k} \left(\dot{q} + \tau_0 \frac{\partial \dot{q}}{\partial t} \right)$$

$$= \frac{1}{\alpha} \left[\frac{\partial T}{\partial t} + \tau_0 \frac{\partial^2 T}{\partial t^2} \right]$$

$$-\nabla \cdot q'' + \dot{q} = \rho c \frac{\partial T}{\partial t}$$

$$-\nabla \cdot \left[-k \left[\nabla T + \tau_T \frac{\partial \nabla T}{\partial t} \right] + \dot{q} \right] = \rho c \frac{\partial T}{\partial t}$$

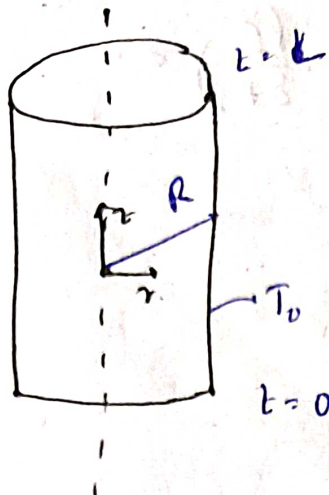
$$= \nabla^2 T + \tau_T \frac{\partial}{\partial t} (\nabla^2 T) + \dot{q}/k = \frac{1}{\alpha} \left(\frac{\partial T}{\partial t} \right)$$

$$-\nabla \cdot q'' + q''' = \rho c \frac{\partial T}{\partial t}$$

$$q = -k \left[\nabla T + \tau_T \frac{\partial \nabla T}{\partial t} \right] - \tau_0 \frac{\partial q}{\partial t}$$

$$q'' =$$

Q6)



$$k = k_0 (1 + \beta_1 T + \beta_2 T^2)$$

$$\nabla(k \nabla T) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$

2-D is heat conduction

$$\frac{\partial T}{\partial t} = 0$$

no heat generation; $\dot{q} = 0$

$$\left[\frac{1}{r} \frac{\partial}{\partial r} \left(-k r \frac{\partial T}{\partial r} \right) + \frac{\partial}{\partial z} \left(-k \frac{\partial T}{\partial z} \right) \right] = \rho c_p \frac{\partial T}{\partial t} = 0$$

$$\frac{1}{r} \frac{\partial}{\partial r} \left(k r \frac{\partial T}{\partial r} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) = 0$$

$$k = k_0 (1 + \beta_1 T + \beta_2 T^2)$$

$$\theta(r, z) = \frac{1}{k_0} \int_{T_0}^{T(r, z)} k(z) dz$$

$$\frac{\partial T}{\partial r} = \frac{\partial \theta}{\partial r} \cdot \frac{\partial T}{\partial \theta} = \frac{k_0}{k(z)} \cdot \frac{\partial \theta}{\partial r}$$

$$\frac{\partial T}{\partial z} = \frac{\partial \theta}{\partial z} \cdot \frac{\partial T}{\partial \theta} = \frac{k_0}{k(z)} \cdot \frac{\partial \theta}{\partial z}$$

$$\frac{1}{r} \frac{\partial}{\partial r} \left(k \cdot r \cdot \frac{\partial \theta}{\partial r} \right) + \frac{\partial}{\partial t} \left(k \cdot \frac{\partial \theta}{\partial t} \right) = 0$$

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r k \frac{\partial \theta}{\partial r} \right) + \frac{\partial}{\partial t} \left(k \frac{\partial \theta}{\partial t} \right) = 0$$

$$k_0 \quad \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \theta}{\partial r} \right) + \frac{\partial \theta}{\partial t} = 0$$

$$@ \quad r=0; \quad \theta = \text{finite}$$

$$r=R; \quad \theta = \theta_0$$

$$@ \quad t=0; \quad -k \frac{\partial \theta}{\partial r} = 0$$

$$t=L; \quad -k \frac{\partial \theta}{\partial t} = \text{const}$$

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \theta}{\partial r} \right) + \frac{\partial \theta}{\partial t} = 0$$

$$@ \quad r=0; \quad \theta = \text{finite}$$

$$r=R; \quad \theta = \theta_0$$

$$@ \quad t=0; \quad -k \frac{\partial \theta}{\partial r} = 0$$

$$t=L; \quad -k \frac{\partial \theta}{\partial t} = \text{const}$$

$$\theta' = \theta - \theta_0$$

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \theta'}{\partial r} \right) + \frac{\partial \theta'}{\partial t} = 0$$

$$@ \quad r=0; \quad \theta', \text{ finite}$$

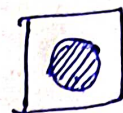
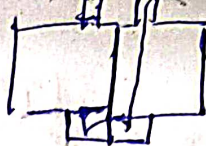
$$r=R; \quad \theta' = 0$$

$$@ \quad t=0; \quad -k \frac{\partial \theta'}{\partial r} = 0$$

$$t=L; \quad -k \frac{\partial \theta'}{\partial t} = \text{const}$$

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \theta'}{\partial r} \right) + \frac{\partial \theta'}{\partial t} = -\lambda$$

$$\frac{1}{r} \frac{\partial^2 \theta'}{\partial r^2} + \frac{\partial \theta'}{\partial r} - \lambda \theta' = 0 \quad (\text{Euler Eqn})$$



~~$$\theta' = A r^\lambda + B r^{-\lambda}$$~~

$$\theta' = A r^\lambda + B r^{-\lambda}$$

$$\theta' = A r^\lambda$$

$$\frac{\partial^2 \theta'}{\partial r^2} = \theta' - \lambda$$

$$\theta' = R(r) Z(z)$$

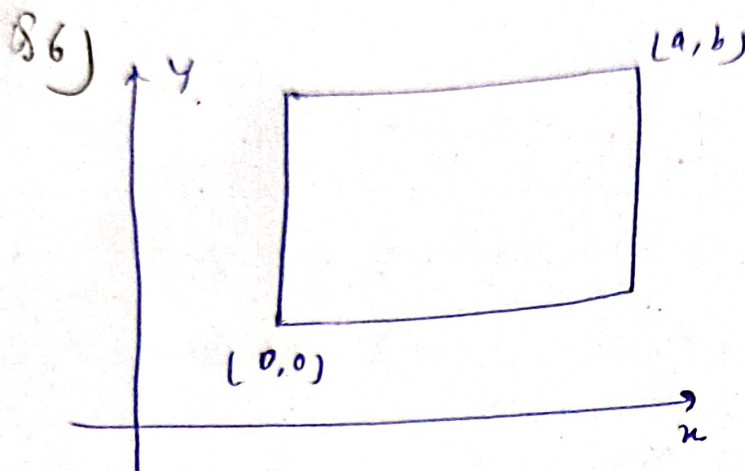
$$\frac{1}{R} \frac{1}{r} \frac{d}{dr} \left(r \frac{dR}{dr} \right) + \frac{1}{Z} \frac{d^2 Z}{dz^2} = 0$$

$$\frac{1}{R} \frac{1}{r} \frac{d}{dr} \left(r \frac{dR}{dr} \right) = -\frac{1}{Z} \frac{d^2 Z}{dz^2} = \lambda$$

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{dR}{dr} \right) = -\lambda$$

$$\frac{d^2 R}{dr^2} + \frac{dR}{dr} + \lambda \cdot r = 0$$

$$\frac{\partial T}{\partial t} = \frac{\partial T}{\partial x} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y}$$



@ $t=0$; $T=T_0$

@ $x=0$; $T=0$

@ $y=0$; $T=0$

$x=a$; $-k \frac{\partial T}{\partial x} = h(T_0 - T_0)$

$y=b$; $-k \frac{\partial T}{\partial y} = h(T_0 - T_0)$

$T_0 = 0$

$\nabla(k \nabla T) + \dot{q} = \rho C_p \frac{\partial T}{\partial t}$

~~no~~ no heat generation ; $\dot{q}=0$

$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$

@ $t=0$; $T=T_0$

@ $x=0$; $T=0$

$x=a$; $-k \frac{\partial T}{\partial x} = h T_0$; $\frac{\partial T}{\partial x} + Bi_0 = 0$

@ $y=0$; $T=0$

$y=b$; $-k \frac{\partial T}{\partial y} = h T_0$; $\frac{\partial T}{\partial y} + Bi_0 = 0$

$$T, X(x)Y(y) \in C^2$$

$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} = \frac{1}{\alpha} \cdot \frac{1}{\tau} \frac{dT}{dt}$$

$$@ x=0; T=T_0$$

$$@ x=a; X=0$$

$$x=a; \frac{dX}{dx} + Bi_0 = 0$$

$$@ y=0; Y=0$$

$$y=b; \frac{dY}{dy} + Bi_b = 0$$

$$\frac{1}{X} \frac{d^2 X}{dx^2} = \frac{1}{\alpha} \cdot \frac{1}{\tau} \frac{dT}{dt} - \frac{1}{Y} \frac{d^2 Y}{dy^2} = -\lambda_1$$

$$\frac{d^2 X}{dx^2} + \lambda_1 X = 0$$

$$@ x=0; X=a$$

$$x=a; \frac{dX}{dx} + Bi_0 = 0$$

$$X_n = C_1 \sin(\lambda_n x)$$

$$\lambda_n \Rightarrow$$

$$\lambda_1 \cot \lambda_1 a = -Bi_0 \text{ (transcendental Eqn)}$$

$$\frac{1}{\alpha} \cdot \frac{1}{\tau} \frac{dT}{dt} - \frac{1}{Y} \frac{d^2 Y}{dy^2} = -\lambda_1$$

$$\frac{1}{\alpha} \cdot \frac{1}{\tau} \frac{dT}{dt} + \lambda_1 = \frac{1}{Y} \frac{d^2 Y}{dy^2} = -\lambda_2$$

$$\frac{dy}{dy} + \lambda_2 y = 0$$

$$@ y=0; y=0$$

$$@ y=b; \frac{dy}{dy} + \lambda_2 y = 0$$

$$y = C_1 \sin(\lambda_1 x)$$

$$\lambda_1 n \rightarrow \lambda_1 \cos \lambda_1 b = -\lambda_2 b$$

$$\frac{1}{\alpha} \cdot \frac{1}{\beta} \frac{dz}{dt} + \lambda_1 + \lambda_2 = 0$$

$$\frac{dz}{z} = -\alpha(\lambda_1 + \lambda_2) dt$$

$$\ln z = C_2 e^{-\alpha(\lambda_1 + \lambda_2) t}$$

$$z(t) = C_3 e^{-\alpha(\lambda_1 + \lambda_2) t}$$

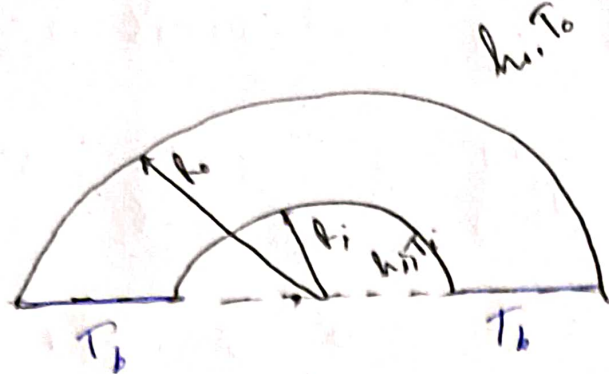
B₁

$$T = \sum_n C_n \sin(\lambda_1 x) \sin(\lambda_2 y) \cdot e^{-\alpha(\lambda_1 + \lambda_2) t}$$

$$@ t=0 \quad T=T_0$$

$$C_n T_0$$

87)



$$\nabla(k \nabla T) + \dot{q} = \rho c \frac{\delta T}{\delta t}$$

$$\frac{1}{r} \frac{\partial}{\partial r} \left(k r \frac{\delta T}{\delta r} \right) + \frac{1}{r^2} \frac{\partial}{\partial \theta} \left(k \frac{\delta T}{\partial \theta} \right) + \dot{q}/k = \frac{1}{\alpha} \frac{\delta T}{\delta t}$$

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\delta T}{\delta r} \right) + \frac{1}{r^2} \frac{\partial^2 T}{\partial \theta^2} = \frac{1}{\alpha} \frac{\delta T}{\delta t}$$

$$\text{@ } r = R_i ; -k \frac{\delta T}{\delta r} = h_i (T_b - T_i)$$

$$\text{@ } r = R_o ; -k \frac{\delta T}{\delta r} = h_o (T_b - T_o)$$

$$T|_{\theta=0} = T|_{\theta=\pi} = T_b$$

$$\text{@ } \theta = 0 ; T = T_b$$

$$\theta = \pi ; T = T_b$$

considering ss;

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\delta T}{\delta r} \right) + \frac{1}{r^2} \frac{\partial^2 T}{\partial \theta^2} = 0$$

$$\text{@ } r = R_i ; -k \frac{\delta T}{\delta r} = h_i (T_b - T_i)$$

$$r = R_o ; -k \frac{\delta T}{\delta r} = h_o (T_b - T_o)$$

$$\left. \begin{array}{l} \text{@ } \theta = 0 ; T = T_b \\ \theta = \pi ; T = T_b \end{array} \right\}$$